

URL Shortener — Deployment Guide

Project: URL Shortener

Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic

Location: /home/dominic/Documents/fastapiurl

Report #: 13/20

Local Development

```
uvicorn main:app --reload --port 8000
```

Production

- Run with `gunicorn -k uvicorn.workers.UvicornWorker main:app`
 - Use nginx as reverse proxy
 - Set up systemd service
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Deployment Guide

Prerequisites

- Python 3.10+
- pip dependencies installed
- Render.com account (or alternative hosting)
- Git repository access

Local Development

```
# Clone repository
git clone https://github.com/voicenotesite/FastAPI-url.git
cd FastAPI-url

# Install dependencies
python -m venv venv
source venv/bin/activate
pip install -r requirements.txt

# Configure environment
cp .env.example .env
# Edit .env with your settings
```

```
# Run server
uvicorn app.main:app --reload
```

Docker Deployment

```
# Example Dockerfile
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 8000
CMD ["uvicorn app.main:app"]
```

Render.com Deployment

1. Create new Web Service in Render dashboard
2. Connect GitHub repository
3. Configure:
4. Build Command: `pip install -r requirements.txt`
5. Start Command: `uvicorn app.main:app`
6. Environment: Python 3.12
7. Region: Frankfurt (Europe)
8. Add environment variables in Render dashboard
9. Deploy (first deploy may take 2-3 minutes)
10. Service auto-deploys on push to main branch

GitHub Pages Deployment

- Used for: WebBartosz, Reports
- Auto-deploys from master branch
- No build step required (static files)
- Custom domain possible via CNAME file

Health Checks

- Endpoint: `GET /health`
- Expected: `{"status": "ok"}`
- Render uses this for service monitoring
- WebBartosz pings all services every 60 seconds

Cold Start Mitigation

- Render free tier spins down after 15 min idle
- First request after idle: 30-60s delay
- Mitigation: WebBartosz pings every 60s keep services warm